Group Number:
Student Names:
Marks: 4 points

COMP 251: Data Structures & Algorithms
Group Assignment 4: Stacks, Queues and Deques

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*********** INSTRUCTIONS ***************

- For **Questions 1 and 2**, assume that the data structures are initially empty and that all operations are performed sequentially.
- For **Question 3**, complete the Java code using `ArrayDeque` and ensure that the output matches the specified format exactly.
- Clearly indicate and briefly justify any modifications made to the `Pair` class in **Question 3**.
- All submitted work must be your own (or your group's) original work. Academic integrity policies apply.

Question 1: Stack operations

What values are returned during the following series of stack operations, if executed upon an initially empty stack? `push(5)`, `push(3)`, `pop()`, `push(2)`, `push(8)`, `pop()`, `pop()`, `push(9)`, `push(1)`, `pop()`, `push(7)`, `push(6)`, `pop()`, `pop()`, `push(4)`, `pop()`, `pop()`.

Question 2: Queue operations

What values are returned during the following sequence of queue operations, if executed on an initially empty queue? `enqueue(5)`, `enqueue(3)`, `dequeue()`, `enqueue(2)`, `enqueue(8)`, `dequeue()`, `dequeue()`, `enqueue(9)`, `enqueue(1)`, `dequeue()`, `enqueue(7)`, `enqueue(6)`, `dequeue()`, `dequeue()`, `enqueue(4)`, `dequeue()`, `dequeue()`.

Question 3: Implementation of Deque

Complete the following code fragment written in Java and in which a class `java.util.ArrayDeque` is imported to implement `Deque`. Make modifications in the `Pair` class (uploaded on the Brightspace) so that you should be able to get the following output

```java

import java.util.ArrayDeque;
public class ArrayDequeDemo
{
   public static void main (String[] args){
```

```java
        ArrayDeque<Pair> a= new ArrayDeque<>();

      // Create objects P1 to P4 of type "Pair". Remember, we
     // studied "Pair" class during one of the earlier lectures.

        a.add(P1);
        a.addFirst(P2);
        a.addFirst(P3);
        a.addLast(P4);

        System.out.println("The contents of Deque are: " + a);
    }
}
```

Output:

The contents of Deque are: [(Laptops,2), (Mobiles,3), (Books,4), (Boards,5)]